**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 093 · Integrated Development Environments

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does a compiler normally do?

Q2. Merge sorted [2, 7] and [4, 6]. Which result is ascending?
A. [2, 4, 6, 7]
B. [2, 7, 4, 6]
C. [4, 6, 2, 7]
D. [7, 6, 4, 2]

Q3. “A constant is a variable whose value must change after every instruction.” Correct this. Explain why.

Q4. A bill is quantity × unit price + fee. “Calculations always run left to right.” Correct this. How does operator precedence affect the result?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

IDE — Integrated Development Environment: software bringing programming tools together.

Editor — Tool used to write and modify source code.

Error diagnostic — A message that helps identify an error and often its location.

Run-time environment — Facility for executing and testing a program.

Translator — Tool that turns source code into executable instructions.

Syntax highlighting — Editor feature that displays different code elements in different colours to help reading.

## New learning

Editor: The editor is where programmers write and change source code. Syntax highlighting and suggestions may make code easier to read and reduce typing mistakes.

Error diagnostics: Diagnostics report issues in code, such as missing brackets or incorrect syntax. A programmer can use the message and line number to find and correct the error.

Run-time environment: The IDE provides a way to run and test the program and inspect the output. This helps programmers check whether the program behaves as expected.

Translators: A compiler or interpreter provided within an IDE converts source code into executable instructions. Together, editor, diagnostics, translator and running tools support the development cycle.

## Worked example

A pupil uses an IDE to create and test a quiz. Describe FOUR different IDE facilities they could use. [4 marks]

1. The editor lets the pupil enter and change the quiz source code [1].

2. Error diagnostics report code errors, helping the pupil find where corrections are needed [1].

3. A translator converts source code into instructions that can be executed [1].

4. The run-time environment runs the quiz so the pupil can inspect results and test the program [1].





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q2. Error diagnostics identify the syntax problem [1], helping the programmer locate and correct it [1]. The IDE's translator converts corrected source into an executable form [1], allowing it to run [1].

Q3. Run the code and inspect the output [1] to establish where it differs from the expected value [1]. Step through the calculations or examine intermediate variable values [1], then correct the faulty line and run again [1].

## Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A compiler translates code before it runs. The executable can run again without translating unchanged source code.

Q2. A — [2, 4, 6, 7]

Q3. A named value that stays the same while the program runs.

Q4. Multiplication comes before addition, giving 10 in the original example. Operator precedence affects the result. Use brackets to make the intended order clear.

### I Do / We Do pages · Guided practice

Q1. Syntax highlighting makes different code elements easier to distinguish [1], making code easier to read [1]. Auto-completion suggests names or commands [1], which can reduce typing errors [1].

Q2. The run-time environment allows a program to be executed and tested [1], letting the programmer inspect behaviour and output [1]. A translator converts source code into a form the CPU can execute [1], enabling the program to run [1].

Q3. Diagnostics identify the possible issue [1] and help locate the line [1]. The editor lets the programmer change the code [1], after which they can run or test it again to check the correction [1].

### You Do pages · Code First

```
total = 5
total = total + 2
print(total)
```

Expected result: 7

### You Do pages · Independent questions

Q1. The editor lets the learner write and change source code [1], so they can improve the quiz instructions [1]. The run-time environment lets them execute the application [1], so they can check that the questions and scores behave correctly [1].







## We Do · Practise with guidance

**Guided 1**

Explain how an editor's syntax highlighting and auto-completion can help a programmer. [4 marks]

Worked method: Syntax highlighting makes different code elements easier to distinguish [1], making code easier to read [1]. Auto-completion suggests names or commands [1], which can reduce typing errors [1].

**Guided 2**

Describe the purpose of the run-time environment and translator provided by an IDE. [4 marks]

Worked method: The run-time environment allows a program to be executed and tested [1], letting the programmer inspect behaviour and output [1]. A translator converts source code into a form the CPU can execute [1], enabling the program to run [1].

**Guided 3**

An IDE reports 'missing closing bracket' at line 2. Explain how diagnostics and the editor help fix the problem. [4 marks]

Worked method: Diagnostics identify the possible issue [1] and help locate the line [1]. The editor lets the programmer change the code [1], after which they can run or test it again to check the correction [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Remove the unintended extra update so the program prints 7.

```
total = 5
total = total + 2
total = total + 2
print(total)
```

iterate

### Question 1

A learner writes and improves a quiz application. Explain how an IDE's editor and run-time environment help during development. [4 marks] Source: Original OCR J277-style practice question.

### Question 2

A programmer cannot run their new source code because it contains a syntax error. Explain how TWO IDE facilities help them get the program running. [4 marks] Source: Original OCR J277-style practice question.

### Question 3

A program should display 10 but displays 12. Describe TWO actions the programmer could perform using their IDE to investigate and fix this logic error. [4 marks] Source: Original OCR J277-style practice question.



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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

An IDE's run-time environment helps a programmer execute and test code.